

VALOQ TERMINAL — COMPLETE INFRASTRUCTURE, VENDOR & GATEWAY MANIFEST

Production: valoq.co.uk | API: api.valoq.co.uk | Search Action Button & Multi-Currency Engine | Next.js 14, FastAPI, NGINX, Supabase, Redis, Gemini 1.5

Platform / Tech	Architectural Role	Why We Used It	Actions, Configurations & Modifications Applied
Porkbun porkbun.com	<i>Domain Registrar & Edge DNS</i>	Cost-effective domain registry with flat renewal pricing and free WHOIS privacy.	- Registered domain: valoq.co.uk. A Record: @ → 76.76.21.21. CNAME: www → cname.vercel-dns.com. CNAME: api → valoq-backend.onrender.com.
Vercel vercel.com	<i>Frontend Edge CDN</i>	Zero-config serverless Next.js edge platform with automated SSL generation.	- Bound custom domain: valoq.co.uk. - Built exchange disambiguation selector badges. - Integrated explicit Search Button & responsive keyboard listeners.
Render render.com	<i>Backend Compute & API Host</i>	Native Python cloud hosting running FastAPI with custom domain routing.	- Web Service: valoq-backend on api.valoq.co.uk. - Implemented dynamic resolve_currency_symbol routing engine (\$, £, ■, €, ¥).
NGINX nginx.org	<i>Reverse Proxy & Rate Limiter</i>	Edge gateway managing request routing, Gzip compression, rate limiting, and SSE passthrough.	- Upstream pools for frontend (:3000) and backend (:8000). - IP rate limits: 30r/s on general API, 5r/s on AI stream. - proxy_buffering off for Gemini streaming.
Docker & Compose docker.com	<i>Containerization Engine</i>	Guarantees identical multi-stage runtime environments across local development and production VPS.	- Node 18 Alpine multi-stage Dockerfile for Next.js. - Slim Python 3.11 Dockerfile for FastAPI. - Unified docker-compose.yml on bridge network valoq-net.
Supabase supabase.com	<i>PostgreSQL & Cold Storage</i>	PostgreSQL with instant SQL execution, indexed schemas, and relational integrity.	Schemas: watchlist, equity_prices (OHLCV candles), equity_financials (normalized statements).
Upstash upstash.com	<i>Serverless Hot-Tier Redis</i>	Fast in-memory cache to prevent repeated upstream queries and bypass rate caps.	- Dual-tier cache engine (Upstash Redis + in-memory fallback). 60s TTL for real-time market quotes and 300s TTL for global search queries.
Google AI Studio aistudio.google.com	<i>Gemini 1.5 Flash Engine</i>	Fast LLM inference for executive memos, SEC 10-K risk audits, and Bull/Bear models.	- Wired /api/v1/stock/ai-stream with Server-Sent Events (SSE). - Real-time typewriter consumption on frontend via ReadableStream.
Yahoo Finance yfinance API	<i>Global Market Feed</i>	Telemetry feed covering global equities, mutual funds, ETFs, commodities, and F&O; instruments.	- Live quote parser and universal search with direct Yahoo Query API fallback. - Real-time 52-week high/low ranges, calendar earnings, and valuation ratios.
UptimeRobot uptimerobot.com	<i>Keep-Alive Ping Monitor</i>	Prevents free/starter cloud instances from hibernating after inactivity.	- Automated HTTPS monitor pinging https://valoq-backend.onrender.com/ every 5 minutes. - Configured real-time alerts for server timeouts.
GitHub github.com	<i>CI/CD Automation & VCS</i>	Central source repository triggering automated deployment pipelines on every commit.	- Monorepo structure (apex-frontend, apex-backend, nginx). - Automated webhooks on main trigger Vercel Edge builds and Render deployments.